Lecture Notes/Summary for Unit 1: Induction

These notes approximately summarize what was covered in lecture. They do not go into heavy detail, but go over the most important points which were brought up in lecture. These notes are not a replacement for going to lecture!

For a lot of the results, they are equivalent to result seen in the textbook, which usually provides them in more detail than these notes.

New! Some material in these notes was not brought up in lecture due to time constraints. This material is explicitly labeled with the "New!" tag you can see to the left.

Syllabus Summary

See the syllabus slides.

Also, please read the full syllabus as posted on Quercus.

Induction

Proving statements of the form $\forall n \in \mathbb{N}.P(n)$ can be difficult. Because natural numbers can have so many different properties, the same proof that works for one number might not work for all.

What does often work, however, is starting from a statement that is true and just slightly tweaking it. While it might often be hard to prove $P(n)$ directly, it often becomes easier if you assume $P(n-1)$. Going from $P(n)$ to $P(n+1)$ is likewise often easier than proving $P(n)$ ex nihilo.

By the transitive property, these implications then cascade. If $P(1) \Rightarrow P(2)$, $P(2) \Rightarrow P(3)$ and so on, then $P(1)$ would imply $P(n)$ for every greater n.

This thus leads to the principle of induction: By proving $P(0)$ and the above implication, you can thus deduce that $P(n)$ holds for every $n \in \mathbb{N}$.

Base Case:	$P(0)$
Inductive Step:	$\forall n \in \mathbb{N}.P(n) \Rightarrow P(n+1)$
Conclusion:	$\forall n \in \mathbb{N}.P(n)$

To properly do a proof by induction, we recommend following the following structure:

1. Define your **predicate** $P(n)$, what you seek to prove for all natural numbers.
2. Prove your **base case** (or base cases) $P(0)$.
3. Make your **inductive hypothesis**: "Let $n \in \mathbb{N}$ be arbitrary, and assume that $P(n)$ is true".
4. Proceed with the **inductive step**, proving that $P(n+1)$ holds given the above assumption. From this, you can then conclude that $\forall n \in \mathbb{N}.P(n) \Rightarrow P(n+1)$.
5. Conclude by **induction** that $P(n)$ holds for all natural n.

Example 1: Consider the sum of all natural nummbers up to n: $0 + 1 + 2 + \cdots + n$. We wish to prove that this is equal to $\frac{n(n+1)}{2}$ for all n.

Let us use induction, and follow the steps above.

1. Define $P(n) : 0 + 1 + \cdots + n = \frac{n(n+1)}{2}$.
2. To prove our base case, observe that $0 = \frac{0 \cdot 1}{2}$, so $P(0)$ trivially holds.
3. **Inductive Hypothesis (IH):** Let $n \in \mathbb{N}$ be arbitrary, and assume that $P(n)$ is true.

   In other words, assume that $0 + \cdots + n = \frac{n(n+1)}{2}$.
4. We wish to now prove $P(n+1) : 0 + \cdots + (n+1) = \frac{(n+1)(n+2)}{2}$. We will prove this via induction.

$$\begin{aligned} 0 + \cdots + (n+1) &= 0 + \cdots + n + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) && \text{(by IH)} \\ &= \frac{(n+2)(n+1)}{2} \end{aligned}$$

   Thus, we see that $P(n+1)$ holds. By generalization, $\forall n \in \mathbb{N}.P(n) \Rightarrow P(n+1)$.
5. By induction, we thus conclude that $\forall n \in \mathbb{N}.P(n)$.

Example 2: A similar proof can be made for the sum of the first n odd natural numbers: $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

Inductive Definition of the Natural Numbers

To justify why induction works, we will define the natural numbers formally. While there are multiple definitions of the natural numbers, we will use a particular one that follows the principle of induction closely, and which we will refer to as the inductive definition of the natural numbers.

Theorem 1. *Let S be a set satisfying the following properties:*

- $0 \in S$,
- $\forall x \in S$, $x + 1 \in S$.

Then, $\mathbb{N} \subseteq S$.

$\mathbb{N}$ is then defined as precisely the minimal set satisfying the conditions in the theorem above. Note that if **New!** both S and S' satisfy the conditions in the above theorem, then so does $S \cap S'$. As such, "the minimal set satisfying the conditions above" is well-defined.

This theorem turns out to be very useful for justifying induction. Given a predicate, $P(n)$, we can define $A = \{n \in \mathbb{N}.P(n)\}$. Then proving $P(0)$ and that $\forall n \in \mathbb{N}.P(n) \Rightarrow P(n+1)$, is equivalent to proving that $0 \in A$ and that $n \in A \Rightarrow n + 1 \in A$. As such, we will have proven that $\mathbb{N} \subseteq A$, which proves that $P(n)$ holds for all $n \in \mathbb{N}$.

Strong Induction

Induction works well for a lot of problems, but runs into issues when the proof of $P(n+1)$ relies on more than $P(n)$.

To fix this, we have the principle of strong induction.

Base Case:	$P(0)$
Inductive Step:	$\forall n \in \mathbb{N}.(\forall k \leq n.P(k)) \Rightarrow P(n+1)$
Conclusion:	$\forall n \in \mathbb{N}.P(n)$

Note that this is very similar to induction, and follows the same proof structure. The only catch is that you may make a stronger inductive hypothesis, assuming not just $P(n)$ but also $P(n-1), P(n-2), \ldots, P(0)$.

Example 3: Consider the Fibonacci numbers, where each term is the sum of the previous two. We can define them as follows:

$$\begin{aligned} F_0 &= 0 \\ F_1 &= 1 \\ F_n &= F_{n-1} + F_{n-2}, \text{ if } n \geq 2 \end{aligned}$$

We wish to prove that if $\Phi = \frac{1+\sqrt{5}}{2}$ and $\phi = \frac{1-\sqrt{5}}{2}$, then $F_n = \frac{\Phi^n - \phi^n}{\sqrt{5}}$ for all n.

To prove this, we define the predicate $P(n) : F_n = \frac{\Phi^n - \phi^n}{\sqrt{5}}$.

Now, let us prove the inductive step first. Let us try first to use simple induction, so we will assume $P(n)$ and try to prove $P(n+1)$.

$$\begin{aligned} F_{n+1} &= F_n + F_{n-1} \\ &= \frac{\Phi^n - \phi^n}{\sqrt{5}} + F_{n-1} && \text{(by IH)} \\ &= \frac{\Phi^n - \phi^n}{\sqrt{5}} + \dots \end{aligned}$$

And here we run into an issue: Because we only assumed $P(n)$, we know nothing about F_{n-1}. In order to make progress, we needed to assume $P(n-1)$, so we should use strong induction.

Proof. Define the predicate $P(n) : F_n = \frac{\Phi^n - \phi^n}{\sqrt{5}}$.

Let us use strong induction. Assume $P(n), P(n-1), \dots, P(0)$.

$$\begin{aligned} F_{n+1} &= F_n + F_{n-1} \\ &= \frac{\Phi^n - \phi^n}{\sqrt{5}} + \frac{\Phi^{n-1} + \phi^{n-1}}{\sqrt{5}} && \text{(by IH, } P(n), P(n-1)) \\ &= \frac{\Phi^{n-1}(\Phi+1) - \phi^{n-1}(\phi+1)}{\sqrt{5}} \\ &= \frac{\Phi^{n-1}\Phi^2 - \phi^{n-1}\phi^2}{\sqrt{5}} \\ &= \frac{\Phi^{n+1} - \phi^{n+1}}{\sqrt{5}} \end{aligned}$$

From this, we see that the inductive step holds.

Now, we need our base case. Proving $P(0)$ is trivial by the observation that $F_0 = 0 = \frac{1-1}{\sqrt{5}}$.

However, observe that our inductive step used both $P(n)$ and $P(n-1)$. And note that when $n = 0$, $P(n-1)$ is not well-defined, and not part of our inductive hypothesis. As such, while we can use our inductive step to prove $P(2), P(3), \dots$, it cannot prove $P(1)$. As such, we need to prove it as a separate base case.

Thankfully, $P(1)$ is also trivial. $\Phi - \phi = \sqrt{5}$, so $F_1 = 1 = \frac{\Phi - \phi}{\sqrt{5}}$.

Thus, we have our base cases and inductive step, and $\forall n \in \mathbb{N}.P(n)$ holds. □

Note that the above proof required several base cases. This often applies with strong induction, where multiple base cases are needed since the inductive step requires multiple prior terms.

Base Cases:	$P(0), P(1), \dots, P(m)$
Inductive Step:	$\forall n \geq m.(\forall k \leq n.P(k)) \Rightarrow P(n+1)$
Conclusion:	$\forall n \in \mathbb{N}.P(n)$

To see how to determine what base cases you need, look at the following example.

New!

Example 4: Suppose we have two types of coins, worth 4 cents and 7 cents, respectively. Note that not all totals can be made with only these coins. For example, you cannot make a total of 9 cents out of these coins.

However, it turns out that these difficult totals are rare. In fact, you can make any total starting at 18 cents!

To prove this, let $P(n)$ be that you can form a total of n cents using only 4 cent and 7 cent coins. Formalizing this more mathematically, we can define $P(n) : \exists a, b \in \mathbb{N}.n = 4a + 7b$.

To determine how many base cases we need, let us start with our inductive step. For our inductive step, let $n \in \mathbb{N}$ be sufficiently large, and assume $P(18), \ldots, P(n)$. We wish to prove that $n + 1$ can be made using the totals above.

Note that by assuming $P(n-3)$, we know that $n - 3 = 4a' + 7b'$ for some $a', b' \in \mathbb{N}$. But if we then let $a = a' + 1$ and $b = b'$, then $4a + 7b = 4(a' + 1) + 7b' = 4a' + 7b' + 4 = n - 3 + 4 = n + 1$. As such, we see that $P(n + 1)$ holds.

Thus, we have our inductive step. But note that this requires us to assume $P(n - 3)$. At the same time, as we are trying to prove the results for all $n \geq 18$, we thus have our base cases starting at $P(18)$. From the above, we see that our inductive step would work for all n except $18, 19, 20, 21$. These will be our base cases, which we can prove separately.

$$\begin{aligned} 18 &= 1 \cdot 4 + 2 \cdot 7 \\ 19 &= 3 \cdot 4 + 1 \cdot 7 \\ 20 &= 5 \cdot 4 + 0 \cdot 7 \\ 21 &= 0 \cdot 4 + 3 \cdot 7 \end{aligned}$$

These base cases and the above inductive step (which will apply for all $n \geq 21$) then give us the conclusion we want by strong induction.

The Well-Ordering Principle

The Well-Ordering Principle states the following:

Theorem 2 (Well-Ordering Principle)**.** *Any non-empty subset A of $\mathbb{N}$ has a minimum element.*

While seeming trivial, this turns out to be a very useful result for proof techniques.

Example 5: Let us prove that for all natural numbers $n \in \mathbb{N}$, $n^2 + n$ is even.

Proof. We will use a proof by contradiction. Assume for contradiction that there exists a counterexample, that there is some n such that $n^2 + n$ is odd.

Then, we can define the set of all counterexamples: $C = \{n \in \mathbb{N}.n^2 + n \text{ is odd}\}$. As our assumption implies that C is non-empty, the WOP states that C must have a minimum element n'.

Now, consider $n' - 1$.

$$\begin{aligned} (n'-1)^2 + (n'-1) &= n'^2 - 2n' + 1 + n' - 1 \\ &= n'^2 + n' - 2n' \\ &= \text{odd} - \text{even} && (n' \in C) \\ &= \text{odd} \end{aligned}$$

Thus, we see that $n' - 1 \in C$.[1] But $n' - 1 < n'$. Thus, we get a contradiction: We have found an element in C smaller than its minimum element.

As such, by contradiction, our assumption must have been wrong. And thus, there must exist no counterexamples. □

[1] Technically, we could have that $n' = 0$ and that $n' - 1 = -1$, which would mean that $n' - 1 \notin C$ as it would not be a natural number. To prevent this edge case, we can prove that $0^2 + 0 = 0$ which is even, so $0 \notin C$, and thus $n' \neq 0$. The WOP also requires an analogue to induction's base case.

The Equivalence of 3 Techniques

With WOP having a "base case-like" step, it feels very reminiscent of induction. In fact, this is because they are secretly the same technique!

Theorem 3. *The following techniques are equivalent, and imply each other:*

1. *The Principle of Induction,*
2. *The Principle of Strong Induction,*
3. *The Well-Ordering Principle.*

In other words, every one of the principles can be derived by assuming another, and everything that can be proven using one can be proven by one of the others. This also means that strong induction is not actually stronger than induction, just that its inductive hypothesis is stronger.

New! *Proof.* To prove this, we will show that each technique implies the next. By the transitive property, it just suffices to prove circular implication, which will imply the rest. This will involve three separate steps.

1. **Induction $\implies$ Strong Induction:** For this step, we may assume the regular Principle of Induction and attempt to prove the Principle of Strong Induction. In other words, we want to prove that if we have $P(0)$ and $\forall n \in \mathbb{N}.(\forall k \leq n.P(k)) \Rightarrow P(n+1)$, then $\forall n \in \mathbb{N}.P(n)$ follows, without applying Principle of Strong Induction.

   To do this, we define the following helper predicate: $Q(n) : \forall k \leq n.P(k)$.

   First, note that $Q(0)$ is $\forall k \leq 0.P(k)$, which is logically equivalent to $P(0)$. As such, since $P(0)$ holds, $Q(0)$ also holds.

   Now, assume for induction that $Q(n)$ holds for some $n \in \mathbb{N}$. Thus, we are assuming $\forall k \leq n.P(k)$. But as we know that $\forall n \in \mathbb{N}.(\forall k \leq n.P(k)) \Rightarrow P(n+1)$, we can thus deduce $P(n+1)$. By conjunction, $Q(n) \wedge P(n+1) = (\forall k \leq n.P(k)) \wedge P(n+1)$ also holds true. But the above is logically equivalent to $\forall k \leq (n+1).P(k)$, which is the definition of $Q(n+1)$. As such, we have just show our inductive step of $\forall n \in \mathbb{N}.Q(n) \Rightarrow Q(n+1)$.

   Thus, by induction, $\forall n \in \mathbb{N}.Q(n)$. But note that for any n, $Q(n) \Rightarrow P(n)$, so this also proves that $\forall n \in \mathbb{N}.P(n)$, and the Principle of Strong Induction has been proven.

2. **Strong Induction $\implies$ Well-Ordering:** We wish to prove that any non-empty subset of $\mathbb{N}$ must contain a minimum element.

   To prove this, we will actually prove the contrapositive of that statement: A subset of $\mathbb{N}$ without a minimum element must be empty.

   To prove this, let A be a subset of $\mathbb{N}$ without a minimum element, and let $P(n)$ be "$n \notin A$". If we then prove that $\forall n \in \mathbb{N}.P(n)$, then no element in $\mathbb{N}$ lies in A, meaning that A must be empty.

   For our base case, note that 0 cannot lie in A. If it did, it would be the minimum element of A as 0 is the minimal natural number. As A has no minimum element, that means that $0 \notin A$ so $P(0)$ holds.

   Now, suppose for strong induction that $P(0), \ldots, P(n)$ holds for some $n \in \mathbb{N}$. Then consider $n+1$. If $n+1$ were in A, then it would be the minimal element, as $0, \ldots, n \notin A$ by our inductive hypothesis so there would be nothing smaller than $n+1$ in A. Thus, since A has no minimal elements, $n+1 \notin A$, so $P(n+1)$ holds.

   By strong induction, $\forall n \in \mathbb{N}.P(n)$ holds, and the proof is complete.

3. **Well-Ordering $\implies$ Induction:** Suppose we know that $P(0)$ and $\forall n \in \mathbb{N}.P(n) \Rightarrow P(n+1)$ holds. We wish to prove that $\forall n \in \mathbb{N}.P(n)$ holds by using the Well-Ordering Principle.

   Our approach will be to define the set of counterexamples $C = \{n \in \mathbb{N}.\neg P(n)\}$. Suppose for contradiction that C were non-empty. Then there would be some minimum $n' \in C$.

Note that since $P(0)$ is given, that proves that $0 \notin C$, so $n' > 0$. As such, $n' - 1 \in \mathbb{N}$. That means that we can apply our inductive step to get that $P(n'-1) \Rightarrow P(n')$. By applying a contrapositive, we see that $\neg P(n') \Rightarrow \neg P(n'-1)$.

Since $n' \in C$, that means that $\neg P(n')$ holds. By using modus ponens on the above, we get that $\neg P(n'-1)$ also holds. But that implies that $n' - 1 \in C$, which contradicts n' being the minimum element of C.

By contradiction, we thus see that C must be empty. But that proves that $\neg\exists n \in \mathbb{N}.\neg P(n) \equiv \forall n \in \mathbb{N}.P(n)$, exactly as desired.

As we have proven circular implication, these three principles are equivalent. □

Induction Pitfalls

When doing a proof by induction, there are several ways it can go wrong.

- Forgetting a base case.

  Example: Let us "prove" that $\forall n \in \mathbb{N}.2n$ is odd.

  Proof. Consider $2(n+1)$. This is equal to $2n+2$. Note that by induction, we can assume that $2n$ is odd. Then $2(n+1)$ is an odd number plus 2 and is thus also odd, so by induction, this holds for all natural numbers. □

- Proving the Converse, $P(k+1) \Rightarrow P(k)$.

  This happens very often by accidentally reversing your proof: You start with what you want to show, and repeatedly derive statements from it until you derive the inductive hypothesis. Then, you conclude that it must be true by assumption. However, this only works if each step of the proof is reversible, which is not guaranteed with this approach.

  Example: Let us "prove" that $\forall n \in \mathbb{N}.n < 10$.

  Proof. Let $P(n)$ be "$n < 10$". Observe that $P(0)$ holds trivially as $0 < 10$.

  Now, suppose that $P(n)$ holds for some n. We wish to prove that $P(n+1)$ holds. In other words, we wish to show that $n + 1 < 10$. Subtracting 1 from both sides we get $n < 9$. But since $9 < 10$, we see that $n < 10$... which we know to be true by induction.

  Thus, by induction, $\forall n \in \mathbb{N}.n < 10$. □

- Missing Edge Cases.

  It's easy to write a proof where the inductive step is almost correct, but which fails on just a few values of n.

  Example: Let us "prove" that any finite set of cows are all the same color.

  Proof. Let $P(n)$ be "any set of n cows are all the same color".

  Note that $P(1)$ holds trivially, as a set of 1 cow is trivially 1 color (the color of that cow).

  Now, suppose for induction that $P(n)$ holds for some arbitrary n. In other words, any set of n cows are all the same color.

  Consider a set of $n+1$ cows. Number these $c_1, \ldots, c_{n+1}$. Note that $\{c_1, \ldots, c_n\}$ is a set of n cows, so these are all the same color. At the same time, $\{c_2, \ldots, c_{n+1}\}$ is also a set of n cows, so c_2 and c_{n+1} are the same color. Both of these hold by our inductive hypothesis.

  But these sets both contain c_2. Thus, all the cows must be the same color, so $P(n+1)$ holds and the result is true by induction. □

- Confusing $P(k) \Rightarrow P(k+1)$ and $P(k-1) \Rightarrow P(k)$.

  Some books do induction by showing $P(k-1) \Rightarrow P(k)$. Make sure that you don't begin by assuming $P(k)$ according to one style and then end up proving $P(k)$ according to the other, as $P(k) \Rightarrow P(k)$ is not a useful statement to prove.

- Proving $P(k) \Rightarrow P(k+2)$.

  If your proof results in this, consider using strong induction with additional base cases.

- Proving $P(k) \Rightarrow Q(k+1)$.

  This most commonly takes the form of Build-Up Error (see below).

Build-Up Error Build-Up Error is a specific type of mistake that happens with induction over some types of objects, most commonly graphs.

Build-Up Error occurs most often when dealing with predicates of the form "for all structures of size n,...". When attempting to prove an inductive step for such a predicate, Build-Up Error happens by taking an object of size n and growing it to an object of size $n+1$. To demonstrate, consider the following "proof".

Proof. We wish to prove that every 2-regular graph consists of a single cycle. Let $P(n)$ be "every n-vertex 2-regular graph consists of a single cycle".

As the smallest possible 2-regular graph has 3 vertices, our base case is $P(3)$.

Now, suppose that $P(n)$ is true. Then every 2-regular simple graph G on n vertices is a cycle graph. Now, observe that to get to a 2-regular simple graph on $n+1$ vertices, we have to delete an edge in G, add a new vertex, and connect it to the endpoints of the deleted edge. Let us label these endpoints v_1 and v_n, and the new vertex as v'.



As G was a cycle graph, it contains a cycle of the form $v_1, v_2, \ldots, v_n, v_1$, which contains all the edges in the graph. Construct a new cycle, $v_1, v_2, \ldots, v_n, v', v_1$ in G'. This clearly contains every edge in G', so G' must be a cycle graph.

Thus, by induction, all 2-regular simple graphs are cycles. □

The issue comes in the construction not being sufficient. This does not prove that every 2-regular graph consists of a single cycle, it proves that every 2-regular graph which can be constructed this way consists of a single cycle. The issue is that not every 2-regular graph can be constructed by the above mechanism. For example, a graph consisting of 2 disjoint 3-cycles is 2-regular, but cannot be constructed from a 5-vertex 2-regular graph by the above construction.

Techniques for Avoiding Errors The best strategy to avoid errors in induction is to follow the template provided: Explicitly state your predicate, write out your inductive hypothesis in full, and clearly state what $P(n)$ is as you try to prove it.

To avoid Build-Up Error, we recommend the Shrink-Regrow Method: To prove the inductive step for a predicate of the form $P(n)$: "forall objects of size n,...", do the following:

1. Select an arbitrary object of size $n+1$ o.
2. Shrink o to an object of size n o'.
3. Prove that the conditions in the inductive hypothesis apply to o'.
4. Apply the IH to o'.
5. Regrow o' to o.
6. Prove that o satisfies the desired properties by using the fact that o' satisfies them.

Structural Induction

Inductively-Defined Sets

Recall our definition of the natural numbers. This time, we will slightly rephrase it to streamline the definition.

Definition 1. *$\mathbb{N}$ is the smallest set satisfying the following properties:*

- $0 \in \mathbb{N}$,
- $\forall x \in \mathbb{N},\ x + 1 \in \mathbb{N}$.

This kind of structure can be used for other sets as well. For example, we can define the set of parenthetical expressions ("x", "$(y + z)$", "$(x - (y \div x))$", etc.) in a similar way.

Definition 2. *$\mathcal{E}$ is the smallest set satisfying the following properties:*

- *"x", "y", "z"$\in \mathcal{E}$.*
- $\forall e_1, e_2 \in \mathcal{E}. \forall \text{ op} \in \{\text{"}+\text{"}, \text{"}-\text{"}, \text{"}\times\text{"}, \text{"}\div\text{"}\}.\ \text{"}(e_1 \text{ op } e_2)\text{"} \in \mathcal{E}$.

In other words, $\mathcal{E}$ contains the three basic variables "x", "y" and "z", and satisfies a rule that allows you to create more complicated expressions out of simpler ones: Any pair of expressions can be joined with an operator to create a new expression.

We can use these rules to try to create some example elements. In particular, we could prove that the example elements we listed before are all in $\mathcal{E}$.

We can also define the set of all possible trees. But we can define it in one of two ways. We can first try to define it inductively, as seen before.

Definition 3. *$\mathcal{T}$ is the smallest set satisfying the following properties:*

- $\varnothing \in \mathcal{T}$.
- $\forall t_1, t_2 \in \mathcal{T}. \forall n \in \mathbb{N}. Node(n, t_1, t_2) \in \mathcal{T}$.

However, we can also define these recursively. By that, we mean that we can define $\mathcal{T}$ in terms of repeated application of the construction rule.

Definition 4. *$\mathcal{T}$ is defined by the following sets:*

- $\mathcal{T}_0 = \{\varnothing\}$.
- $\mathcal{T}_{i+1} = \{Node(n, t_1, t_2) : n \in \mathbb{N}, t_1, t_2 \in \mathcal{T}_i\} \cup \mathcal{T}_i$.
- $\mathcal{T} = \mathcal{T}_0 \cup \mathcal{T}_1 \cup \mathcal{T}_2 \cup \cdots$.

It turns out that these last two definitions are equivalent, but we will not be going into details with it.

All of these are examples of inductively-defined sets. An inductively-defined set is one that is defined in terms of some basis elements, and a set of construction rules (also known as closure properties).

Specifically, an inductively-defined set is generally in the following form:

> A is defined as the smallest set satisfying the following properties:
>
> 1. Basis: $b_0, b_1, \ldots \in A$
> 2. Construction Rule: $\forall x_1 \in A. \forall x_2 \in A. \cdots \forall x_{k_1} \in A. f_1(x_1, \ldots, x_{k_1}) \in A$
> 3. Construction Rule: $\forall x_1 \in A. \forall x_2 \in A. \cdots \forall x_{k_2} \in A. f_2(x_1, \ldots, x_{k_2}) \in A$
> 4. Construction Rule: $\vdots$

In other words, A must contain some minimum basis elements $b_0, b_1, \ldots$, and then A must be closed under some functions $f_1, f_2, \ldots$.

Structural Induction

We wish to be able to prove things about inductively-defined sets. For this, we can take advantage of the structure given by our inductive definitions.

To demonstrate, we will define a few relevant functions. For an expression $e \in \mathcal{E}$, let $\text{vr}(e)$ represent the total number of variables that appear in e and let $\text{op}(e)$ represent the total number of operations that appear in e. For example, if $e = (x + (y \div z))$, then $\text{vr}(e) = 3$ and $\text{op}(e) = 2$.

We now want to demonstrate a relationship between these two quantities.

Theorem 4. $\forall e \in \mathcal{E}.\, \text{vr}(e) = \text{op}(e) + 1$.

Proof. Let us begin by defining a predicate. Let $P(e)$ be "$\text{vr}(e) = \text{op}(e) + 1$.

First, consider the basis elements of $\mathcal{E}$. These are "x", "y", and "z". For each of these, $\text{vr}(e) = 1$ and $\text{op}(e) = 0$. Thus, $P(e)$ holds for each basis element of $\mathcal{E}$.

Now, let $e_1, e_2 \in \mathcal{E}$ be arbitrary and assume that $P(e_1)$ and $P(e_2)$ hold.

Consider $e =$ "$(e_1 + e_2)$". We can evaluate the number of variables and operations in e.

$$\begin{aligned} \text{op}(e) &= \text{op}(e_1) + \text{op}(e_2) + 1 \\ \text{vr}(e) &= \text{vr}(e_1) + \text{vr}(e_2) \\ &= \text{op}(e_1) + 1 + \text{op}(e_2) + 1 && \text{(By } P(e_1), P(e_2)\text{)} \end{aligned}$$

From this, we thus see that $\text{vr}(e) = \text{op}(e) + 1$, and that thus $P(e)$ holds. A similar argument holds for the other three possible operators as well.

Thus, by structural induction, $\forall e \in \mathcal{E}.P(e)$. □

This last conclusion could be done due to structural induction. Structural induction is a rule of inference analogous to regular induction, applicable to an inductively-defined set.

If we use the same inductively-defined set A as before, structural induction is the following argument:

1. $P(b_0), P(b_1), \ldots$
2. $\forall x_1 \in A. \forall x_2 \in A. \cdots \forall x_{k_1} \in A. (P(x_1) \wedge P(x_2) \wedge \cdots P(x_{k_1})) \Rightarrow P(f_1(x_1, \ldots, x_{k_1}))$
3. $\forall x_1 \in A. \forall x_2 \in A. \cdots \forall x_{k_2} \in A. (P(x_1) \wedge P(x_2) \wedge \cdots P(x_{k_2})) \Rightarrow P(f_2(x_1, \ldots, x_{k_2}))$
4. $\vdots$

$\therefore$ $\forall x \in A. P(x)$

In essence, if you prove that P is true for all basis elements of the set A, and then prove that P is "maintained" under each of the construction rules, then P must be true for all elements of the set.

We can thus use this to prove something for any inductively-defined set, just by following a strategy similar to that for induction:

1. Define your **predicate** $P(x)$, which you seek to prove for all $x \in A$.
2. Prove your **base cases** $P(b_0), P(b_1), \ldots$ for each basis element.
3. Make your **inductive hypothesis**: "Let $x_1, x_2, \ldots, x_{k_1} \in A$ be arbitrary and assume that $P(x_1), \ldots, P(x_{k_1})$ are all true."
4. Proceed with the **inductive step**, prove that $P(f_1(x_1, \ldots, x_{k_1}))$ holds by utilizing the above construction.
5. Repeat the above for every construction rule in your set.
6. Conclude by **structural induction** that $P(x)$ holds for all $x \in A$.

We can justify why this works by defining a new set: $A_P = \{x \in A.P(x)\}$. Observe that the statements "$P(x)$" and "$x \in A_P$" are equivalent by definition. Thus, if we translate the conditions of structural induction, we get the following statements:

1. $b_0, b_1, \ldots \in A_P$
2. $\forall x_1 \in A_P. \forall x_2 \in A_P. \cdots \forall x_{k_1} \in A_P. f_1(x_1, \ldots, x_{k_1}) \in A_P$
3. $\forall x_1 \in A_P. \forall x_2 \in A_P. \cdots \forall x_{k_2} \in A_P. f_2(x_1, \ldots, x_{k_2}) \in A_P$
4. $\vdots$

Note, however, that these statements are exactly the conditions from the definition of A. Note also that $A_P \subseteq A$ by definition. At the same time, if there existed elements in A that are not in A_P, then A_P would be smaller than A, despite satisfying all the conditions in the definition of A. This would contradict the definition of A as the smallest such set, which implies that $A_P = A$. But that then implies that $\forall x \in A.P(x)$, exactly as desired.